LearnLive: An Integrated Platform for Online Coding Education

LearnLive is a web-based platform made to bring all the tools needed for online coding education into one place. WEEPRO Coders faced a big problem because teachers, students, and parents were using many different apps like Zoom, WhatsApp, Google Drive, Replit, and Kahoot. This caused confusion, extra work, communication gaps, and made it hard to track students' progress. Our project solves this by creating a single platform where all major tasks can be done easily and in one system.

LearnLive supports four types of users: Admins, Teachers, Students, and Parents. Teachers can take live coding classes, see and control students' code using a built-in coding editor, upload quizzes and assignments, check submissions, generate certificates, and talk to students and parents through the app. Students can join classes, write and run code, submit homework and quizzes, ask questions in discussion forums, and see their progress. Parents can check their child's attendance, performance, class tasks, certificates, and communicate with teachers. Admins can schedule classes, assign teachers and students to camps, manage permissions, track attendance, and view important analytics like enrollment numbers and retention rates.

The platform is built using the **MERN stack: ReactJS** for the frontend because it is fast and easy to manage, and **NodeJS/Express with MongoDB** for the backend because it works well with real-time features and high traffic. **Socket.IO** is used for live communication in chats and collaborative coding, and the **Judge0 API** is used to run code inside the platform. Cloud hosting makes the system available at all times.

The project was developed in steps, starting with the learning management system, then adding live classes, chat features, collaborative coding, student assessments, and finally the analytics dashboard. Each feature was tested using real use cases such as login, class creation, assignment uploading, certificate generation, and joining collaborative coding rooms.

In the end, LearnLive replaces many separate apps with one complete solution. It makes communication clearer, reduces manual work, helps teachers manage classes better, and gives students and parents a more organized learning experience. It also provides useful data to help WEEPRO Coders plan future programs. Overall, LearnLive offers a simple, efficient, and reliable platform for online coding education.